

Generation Interconnection Request System Impact Study Report

For Queue Project Q20-1616

Zona Libre 6007 230 kV
October 17, 2022

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q20-1616. The Interconnection Customer proposes a 1500.0 MW natural gas combined cycle facility interconnecting at the Zona Libre 6007 230 kV substation in the Ohio area. The request was received on December 06, 2020. The study evaluated system intact and single contingency conditions using the regional planning case, including 407 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 5 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$110,406,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

For the loss of Post Rock Substation - Myrtle Pt 230 KV, the Mcmillan - Myrtle Pt 230 KV loads to 133.6% of its 524 MVA rating. Project DFAX: 22.8%.

The Myrtle Point Substation - George Neal S 230 KV is overloaded to 112.0% of its 1,448 MVA emergency rating for the loss of the Swan Valley - Trego 500 KV Line. The project contributes 64.8% DFAX (972.0 MW) to this overload.

George Neal S - Zona Libre 6007 230 KV Line: 111.6% loading (1,476 MVA) following the outage of the Swan Vly - Trego 500 KV; DFAX = 64.8%.

Eastown Rd - Moreno Tap 230 KV: 106.2% loading (1,345 MVA) following the outage of the Zona Libre 6007 - George Neal South 230 KV; DFAX = 100.0%.

Eastown Road 100-1 kV TX: 104.0% loading (213 MVA) following the outage of the Zona Libre 6007 - George Neal S 230 KV L; DFAX = 8.1%.

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Mcmillan - Myrtle Pt 230 KV	Rebuild line with higher-capacity conductor	\$12,948,000	100% (\$12,948,000)
Myrtle Point - George Neal S 230 KV line	Rebuild line with higher-capacity conductor	\$20,654,000	92% (\$19,101,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
George Neal S - Zona Libre 6007 230 KV Line	Rebuild line with higher-capacity conductor	\$50,577,000	92% (\$46,774,000)
Eastown Rd - Moreno Tap 230 KV	Reconductor line	\$29,015,000	100% (\$29,015,000)
Eastown Road Substation 100/1kV TX	Replace transformer with higher-rated unit	\$3,488,000	74% (\$2,568,000)
	Total allocated to project		\$110,406,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.